

DUST CAN REDUCE ENERGY OUTPUT

In dusty India, PV solar power plants can lose up to 30 per cent energy output within a few weeks of installation

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Photovoltaic (PV) solar panels are today the talk of the town. The cost of solar power is a concern, though it is as much as 20 to 25 per cent less than that from diesel generators. Additional bonus for going solar is elimination of noise and air pollution. More and more people are turning pro-solar, especially because the line power quality continues to be poor with long blackouts and voltage swings. In addition, PV solar prices are dropping in India as competition builds up and China is offering good quality at very competitive prices.

However, there is one problem: PV solar companies rarely warn buyers about the deadly impact of dust and dirt on the PV panel output. As a consequence, few people have realised its gravity. Within just a week, there is a sharp drop in the power output of solar panels due to ever-present dirt in our cities and towns!

Solar panels glisten in the sun after first being installed. But after a little while, these panels become progressively coated with dirt like dust and bird droppings. Flocks of birds leave behind a mess after spending the night overhead while dust, grime and mould add to the layers of dirt, colouring the solar panels a splotchy brown. Suppliers of solar systems, however, often assume that their work is over after the panels are installed and payment received.

Dirt is a major problem with solar panels, and letting it accumulate over a few weeks will reduce a solar panel's



efficacy by almost 30 per cent even in dust-free Europe. It is therefore essential to maintain them with regular cleaning; at least every other day. Fortunately, this is the only maintenance that solar panels need. Therefore when you install a solar power plant, the ease of regularly cleaning the PV panels has to be kept in mind.

There are automatic cleaning systems like SolarWash and others available abroad but manual cleaning is a far more inexpensive option in India. Cleaning thrice a week and washing once a month is essential in most of the Indian locations. If one does not clean regularly, the dirt actually gets bonded to the surface. Just imagine what happens to your car windshield if it's not cleaned daily. PV panels are in open like the car. If you don't keep them very clean, don't expect to get the rated power output specified by the supplier and also the promised return on your investment.

The best example of rather thoughtless application of solar power is pole-mounted solar streetlights. Most of these are bought by public bodies with public money. Like any other public utility, the municipality, the *gram-panchayat* or the city corporation that installs these never bothers to maintain them. Such projects basically start and end with spending the public money and lining the pockets both by the buying officials and the seller. Maintenance and supervision are never given a thought.

A study conducted by two citizen groups shows that 70 per cent of these streetlights stop doing their job within three months of installation. Pole-mounted panels are inaccessible without ladders, so no one bothers to clean or ever check them. ●

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